Parameter	4G (legacy)	5G (current integration phase)	development)	networ
			infrastructure layer	and inte mission
Primary use stage	Legacy operations	Current implementation and scaling	Research, testing, emerging pilot networks	Operati for rem sensing maritim defense expediti

9.7 Space Exploration and Drones

Drones are poised to play a critical role in the future of space exploration, offering mobility, adaptability, and autonomy in environments where traditional rovers, landers, and human astronauts face severe physical and operational constraints. Planetary surfaces such as those on Mars, the Moon, Titan, and asteroids present rugged terrain, thin or non-existent atmospheres, extreme temperature variations, and communication delays that make ground-based navigation slow and risky. Aerial and hopping drones, however, can navigate three-dimensional space, bypass obstacles, and survey wide regions quickly, making them ideal for scientific exploration, environmental mapping, and mission support operations.

Advances in lightweight composite materials, radiation-hardened avionics, autonomous navigation, multispectral imaging systems, and compact power sources are enabling drones to function effectively beyond Earth. Missions such as NASA’s Ingenuity on Mars and the forthcoming Dragonfly mission to Titan demonstrate how drones can expand the scope of planetary science, discovering new features, identifying suitable landing sites, and collecting data that rovers and human teams cannot easily access.